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WORKSHEET 6

AIM: To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyze how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

S/W Requirement:
- Database Management System: PostgreSQL / Oracle Database Express Edition
- Database Administration Tool: pgAdmin

OBJECTIVES:

To implement and analyze the use of:

- Implicit cursors
- Explicit cursors
- Cursor attributes

for processing multiple rows from a database table and applying business logic effectively.

PROBLEM STATEMENT:

In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules.

PRACTICAL/EXPERIMENT STEPS:

Design and execute PL/SQL programs that demonstrate:

- The use of implicit cursors for single-row DML operations
- The use of explicit cursors to fetch and process multiple records
- The application of cursor attributes to control program execution.

1. **PROBLEM STATEMENT – Implicit Cursor**

Write a PL/SQL program to update salary and display the number of rows affected using an implicit cursor.

PROGRAM:

```
BEGIN
    UPDATE Employee_data
    SET salary = salary + 1000
    WHERE emp_id = 101;

    IF SQL%FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Rows Updated: ' || SQL%ROWCOUNT);
    ELSE
        DBMS_OUTPUT.PUT_LINE('No rows updated');
    END IF;
END;
```

OUTPUT:

```
Rows Updated: 1

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.011
```

2. PROBLEM STATEMENT – Explicit Cursor

Fetch and display employee records using an explicit cursor.

PROGRAM:

```
DECLARE
    CURSOR emp_cursor IS
        SELECT emp_id, emp_name, salary FROM Employee_data;
    v_id employee.emp_id%TYPE;
    v_name employee.emp_name%TYPE;
    v_salary employee.salary%TYPE;
BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id, v_name, v_salary;
        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' ' || v_salary);
    END LOOP;

    CLOSE emp_cursor;
END;
```

OUTPUT:

```
101 John 4000
102 Alice 4000
103 Bob 3500
104 Emma 4500


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.107
```

3. PROBLEM STATEMENT –Cursor Attributes

Demonstrate the use of cursor attributes such as %FOUND, %NOTFOUND, and %ROWCOUNT.

PROGRAM:

```
DECLARE
    CURSOR emp_cursor IS
        SELECT emp_id FROM Employee_data;

    v_id employee.emp_id%TYPE;
BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id;

        IF emp_cursor%FOUND THEN
            DBMS_OUTPUT.PUT_LINE('Employee ID: ' || v_id);
        END IF;

        EXIT WHEN emp_cursor%NOTFOUND;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Total Rows Fetched: ' || emp_cursor%ROWCOUNT);

    CLOSE emp_cursor;
END;
```

OUTPUT:

```
Employee ID: 101
Employee ID: 102
Employee ID: 103
Employee ID: 104
Total Rows Fetched: 4


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006
```

LEARNING OUTCOMES:

1. Understood the role of cursors in PL/SQL for handling multi-row query results
2. Learned the difference between implicit and explicit cursors
3. Used cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN
4. Developed PL/SQL programs to process records row by row
5. Applied cursor-based logic in practical database scenarios.

CONCLUSION:

This experiment demonstrated the use of implicit and explicit cursors in Oracle PL/SQL.